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More Than a 2-Fold Risk of Contralateral Anterior Cruciate Ligament Injuries Compared With Ipsilateral Graft Failure 10 Years After Primary Reconstruction

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Background: Failure of anterior cruciate ligament (ACL) reconstruction or an injury to the ACL in the contralateral knee represents a devastating event for patients, especially those young and physically active. However, controversies are still present regarding long-term failure rates and risk factors.

Purpose: To assess the long-term rate of ipsilateral graft failure and contralateral ACL injuries after ACL reconstruction performed at a single center using the same surgical technique with a hamstring autograft and to investigate the effect of sex, age, and pre-injury activity level as predictors of second ACL injuries.

Study Design: Case series; Level of evidence, 4.

Methods: The study cohort consisted of 244 consecutive patients (mean age, 30.7 years) who underwent ACL reconstruction with a single bundle plus lateral plasty technique using the hamstring tendon between November 2007 and May 2009. The number of subsequent ACL injuries (ipsilateral ACL revision or contralateral ACL reconstruction) was determined at a minimum follow-up of 10 years. Survivorship of either knee and subgroup analysis included sex, age, preoperative Tegner activity level, timing of ACL reconstruction, body mass index, and smoking status.

Results: Ipsilateral ACL revision was performed in 8 (3.4%) patients and contralateral ACL reconstruction in 19 (7.8%) patients. Only 1 patient had both ipsilateral and contralateral injuries. No predictors were found for ipsilateral ACL revision, while age $<$18 years and preoperative Tegner level $\geq$7 had a higher risk of contralateral ACL reconstruction. The highest rate of a second ACL reconstruction procedure was in young ($<$18 years) and active (Tegner $\geq$7) patients, in whom the 10-year survival of either knee was 61.1%. Six years after primary ACL reconstruction, the rate of contralateral ACL reconstruction was significantly higher than that of ipsilateral ACL revision (hazard ratio, 2.4-3.6).

Conclusion: In the long term, a second injury to either the ipsilateral or the contralateral knee in young and active populations could reach 40%, with a more than double-fold risk of contralateral ACL reconstruction compared with ipsilateral ACL revision.

Keywords: ACL; knee; failure; ipsilateral; contralateral; second injury

Knee injuries after anterior cruciate ligament (ACL) reconstruction represent devastating events for patients, especially those young and involved in demanding sports. ACL graft failure has been estimated at a rate of around 5.8%[29] in the midterm and 7.9% after 10 years.[11] Its risk factors have been investigated in several prospective and cohort studies, with young age, early return to sport, family history of ACL injuries, and return to strenuous activities being the most relevant.[3,6,8,22-24] However, considerable inconsistencies, especially regarding sex and surgical characteristics, remain.[10,15,17,28] The large number of studies performed in the past decade points out the clinical relevance of graft failure because it represents a serious concern for patients who undergo ACL reconstruction. However, the likelihood of a contralateral ACL injury is often underestimated in the surgeon's mind, despite that its occurrence could have as much of a disastrous impact as primary reconstruction failure.[14] Only a few studies have looked at the incidence and risk factors for contralateral ACL injuries, which are summarized in a systematic review that identified a similar risk of ipsilateral ACL failure (7%) and contralateral ACL injuries (8%).[26] However, several large

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monocentric prospective studies with a long-term follow-up[3,19] suggest a higher incidence of contralateral injuries. Conflicting results have been reported based on patient age, sex, anatomic features, and the graft used for primary ACL reconstruction, as well as follow-up.[3,8,15,18,28] Therefore, the risk of a second ACL injury, either ipsilateral or contralateral, represents a not well-understood subject; the inconsistencies reported in the current literature do not allow adequate patient counseling and the development of effective preventive measures.

Considering this background, the aim of the present study was to assess the long-term risk of ipsilateral graft failure and contralateral ACL injuries after ACL reconstruction performed at a single center using the same surgical technique with a hamstring autograft. Moreover, predictors of a second ACL injury were investigated. The hypotheses were that the rates of second injuries would be similar for ipsilateral and contralateral knees and that it would be possible to identify patients at increased risk for second ACL injuries.

METHODS

Patient Selection

A prospectively collected patient practice database was searched for demographic, injury, and surgical information. Consecutive patients who underwent isolated primary ACL reconstruction between November 2007 and May 2009 in the sport traumatology department of IRCCS Istituto Ortopedico Rizzoli were identified and screened for eligibility. Only primary ACL reconstruction with no other concomitant procedures, with the exception of meniscal treatment, was considered eligible for inclusion. Patients were excluded if they had a previous ipsilateral or contralateral ACL injury or surgery. Medical charts were reviewed to identify further ipsilateral or contralateral ACL reconstruction procedures performed within the considered follow-up. Also, to identify ACL reconstruction performed outside our institution, all the patients were contacted to assess the occurrence of further knee surgical procedures, similarly to other cohort studies.[23,24] Based on medical charts and patient interviews, the following information was obtained: age at surgery, sex, body mass index (BMI), smoking status, preoperative Tegner activity level, time elapsed between injury and surgery, and date of further ipsilateral or contralateral ACL reconstruction. The protocols of this study were approved by the local ethics committee (No. 0005169).

Surgical Procedure

All the patients underwent the same surgical technique of single-bundle over-the-top ACL reconstruction plus lateral plasty using both hamstring tendons.[30] After tendon harvest preserving the tibial attachment, the graft was passed through the tibial tunnel and "over the top" of the femur. A total of 10 cycles of knee flexion-extension was performed for graft pretensioning. A groove was made with an osteotome on the lateral aspect of the lateral condyle of the femur to freshen the bone to enhance healing of the graft to the femur while aiding in graft stabilization. The knee was flexed to 90° with 45° of external tibial rotation, and the posterior drawer test was applied. The tendons were then fixed to the cortical bone of the femur with 2 bone staples under manual maximum tension. The remaining part of the graft was then passed deep to the iliotibial band and superficial to the lateral collateral ligament, then fixed with a single staple onto the Gerdy tubercle by extra-articular plasty with the knee flexed at 90° with 30° of external tibial rotation.

All patients underwent the same postoperative rehabilitation protocol. A knee brace was not used postoperatively. Range of motion, quadriceps muscle active exercises, straight-leg raises, and prone hamstring muscle–stretching exercises were all begun the day after surgery. Functional muscle stimulation was performed for 2 hours, 3 times daily, for 4 weeks. Patients were allowed partial weight-bearing during the first 2 weeks. Furthermore, patients were allowed full passive extension and active flexion through the range from 0° to 120° starting from the third postoperative day. Then, 3 weeks after surgery, full weight-bearing was allowed. Stationary biking, active knee extension with weights, and one-quarter squats were introduced 4 weeks after surgery. Running was started at 3 months and pivoting sport activities after 6 months.

Statistical Analysis

Statistical analysis was performed with MedCalc (MedCalc Software). Continuous variables were reported as the mean ± standard deviation, while categorical variables were reported as the absolute number and proportion of the total sample. Only the Tegner activity level was reported as the median with interquartile range. An independent-samples *t* test was used to compare the continuous variables between all included patients and those lost to follow-up, the Mann-Whitney test to compare Tegner activity levels, and the Fisher exact test to compare dichotomous categorical variables. Kaplan-Meier survival curves were constructed using the time to ipsilateral ACL revision or contralateral ACL reconstruction as the endpoint.

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TABLE 1
Patient Characteristics[a]

	All Patients (N = 244)	Patients Lost to Follow-up (n = 48)	*P* Value
Follow-up, mean ± SD, y	10.0 ± 0.5	10.1 ± 0.5	.2063
Sex			
Male	186 (76)	39 (81)	.5734
Female	58 (24)	9 (19)	
Age, mean ± SD, y	30.7 ± 10.6	30.6 ± 10.4	.9523
≥18 (old)	215 (88)	43 (90)	.5040
<18 (young)	29 (12)	5 (10)	
Preoperative Tegner level, median (IQR)	7 (5-7)	7 (5-7)	.9711
<7 (low)	108 (44)	24 (50)	.5257
≥7 (high)	136 (56)	24 (50)	
Timing of reconstruction, mean ± SD, mo	16 ± 15	18 ± 16	.4043
<3	183 (75)	39 (81)	.4598
≥3	61 (25)	9 (19)	
BMI, mean ± SD, kg/m^2	23.9 ± 3.5	24.2 ± 3.7	.5691
<25	168 (69)	35 (73)	.6120
≥25	76 (31)	13 (27)	
Smoker			
No	169 (69)	31 (65)	.6103
Yes	75 (31)	17 (35)	

[a]Data are presented as n (%) unless otherwise specified. BMI, body mass index; IQR, interquartile range.

The survival rates with 95% confidence intervals (CIs) at 2-, 4-, 6-, 8-, and 10-year follow-up were calculated. To compare the survival curves according to different variables with the log-rank test, the variables were dichotomized as follows: male or female sex, age <18 (young) or ≥18 years (old), BMI <25 (normal weight) or ≥25 kg/m^2 (overweight), preoperative Tegner activity level <7 (low) or ≥7 (high), time to ACL reconstruction <3 (acute) or ≥3 months (chronic), and nonsmoker or smoker. Hazard ratios (HRs) with 95% CIs were calculated as well. Stepwise multiple logistic regression was also performed to assess the relative contribution of age, sex, BMI, smoking status, preinjury Tegner activity level, and timing of ACL reconstruction on the previous outcomes. In the case of significant variables in the regression model, these were combined to calculate the survival rate and HR of patient subgroups. Finally, the HR between the risk of ipsilateral ACL revision or contralateral ACL reconstruction was calculated at the different follow-up points. Statistical significance was set at $P < .05$.

RESULTS

Patients' Demographics

Overall, 318 consecutive patients were considered eligible for the study, and the data of 270 (84.9%) were available for the evaluation of subsequent ipsilateral or contralateral ACL reconstruction. A total of 26 patients (9.6%) were excluded because they had ACL reconstruction performed before the index procedure, leaving 244 patients (186 male; 58 female) with a mean age of 30.7 ± 10.6 years in the final analysis at a mean of 10.0 ± 0.5 years (Figure 1). Preoperative baseline characteristics were similar between all patients included in the analysis and those lost to follow-up (Table 1). There were 8 patients (3.4%) who underwent ipsilateral ACL revision and 19 patients (7.8%) who underwent contralateral ACL reconstruction within the considered follow-up. Only 1 patient (0.4%) underwent both ipsilateral ACL revision and contralateral ACL reconstruction. Isolated traumatic meniscal lesions occurred in 3.3% of ipsilateral knees and 2.4% of contralateral knees.

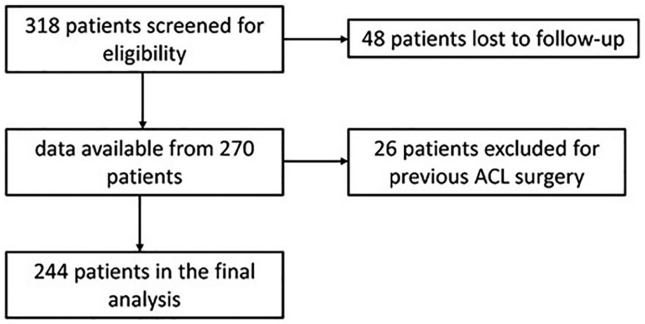

Figure 1. The flowchart shows the patient screening process from the institution database and the inclusion for final analysis. ACL, anterior cruciate ligament.

Ipsilateral ACL Revision

The mean time to ipsilateral ACL revision was 3.7 ± 3.0 years, with an annual rate of 0.34% per year; because many injuries occurred within the first 2 years (38%), the corrected annual rate was 0.61% up to 2 years and 0.26% from 2 to 10 years. Therefore, the 10-year survival from ipsilateral ACL revision was 96.3%, with no differences between male and female patients (Figure 2). Both the logistic regression

TABLE 2
Logistic Regression Analysis of ACL Injury Risk[a]

	Ipsilateral ACL Revision		Contralateral ACL Reconstruction		Second ACL Rupture	
Variables	OR (95% CI)	*P* Value	OR (95% CI)	*P* Value	OR (95% CI)	*P* Value
Age (<18 y)	Excluded from model		4.0 (1.3-12.1)	.0154	3.2 (1.2-8.7)	.0243
Preoperative Tegner level (≥7)	Excluded from model		16.2 (2.1-124.0)	.0075	4.9 (1.6-14.9)	.0047
Overall model fit	NS			.0001		.0004

[a]Sex, acute reconstruction, overweight, and smoker were excluded from the model because $P > .05$. Age <18 years and high activity level were found to be significant predictors for contralateral ACL reconstruction or a second ACL rupture. ACL, anterior cruciate ligament; NS, not significant; OR, odds ratio.

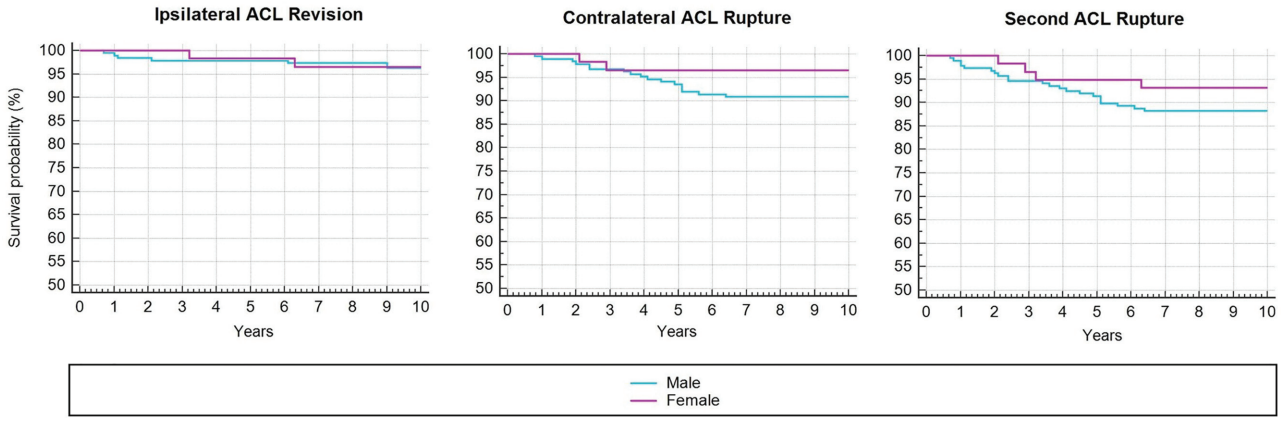

Figure 2. Survival curves of ipsilateral anterior cruciate ligament (ACL) revision, contralateral ACL reconstruction, and a second ACL injury according to sex. No significant differences between sexes were found.

analysis (Table 2) and the univariate analysis (Appendix Table A1, available in the online version of this article) did not identify significant predictors of ipsilateral ACL revision (Figure 3).

Contralateral ACL Reconstruction

The mean time to contralateral ACL reconstruction was 3.5 ± 1.6 years, with an annual rate of 0.78% per year; because most of the injuries occurred within the first 3 years (42%), the corrected annual rate was 1.11% up to 3 years and 0.67% from 3 to 10 years. Therefore, the 10-year survival from contralateral ACL reconstruction was 92.2%, with no differences between male and female patients (Figure 2). In the univariate analysis, only age at surgery <18 years and a preoperative Tegner level ≥7 showed higher HRs of injury (Appendix Table A2, available online). Furthermore, the combination of young age and high activity level had an HR of 32.3, while the 10-year survival was 66.7% in comparison with 99% in patients with age ≥18 years and preoperative Tegner level <7 (Figure 4). The significant role of young age (odds ratio [OR], 4.0; $P = .0154$) and high level of activity (OR, 16.2; $P = .0075$) was confirmed also in the logistic regression model (Table 2).

Second ACL Rupture

Considering the 27 ipsilateral or contralateral ACL ruptures that occurred in 26 patients (10.7%), the 10-year survival from a second ACL rupture was 89.3%. Similar to contralateral injuries, age at surgery <18 years and a preoperative Tegner level ≥7 showed higher HRs of injury in the univariate analysis (Appendix Table A3, available online). Specifically, patients with age <18 years and a preoperative Tegner level ≥7 had a 10-year survival rate of 61.1%, compared with 95.9% in older and less active patients; moreover, they had a significant HR of 12.7 (Figure 5). The significant role of young age (OR, 3.2; $P = .0243$) and high level of activity (OR, 4.9; $P = .0047$) was confirmed also in the logistic regression model (Table 2). Finally, a significantly higher ($P < .05$) risk of contralateral reconstruction compared with ipsilateral revision was noted starting from the sixth postoperative year (Table 3). Specifically, there was a more than 4-fold HR of contralateral injury within the first 6 years in male patients and in patients with a preoperative Tegner level ≥7 (Figure 6).

DISCUSSION

The most important finding of the present study was that a more than two-fold risk of contralateral ACL injuries

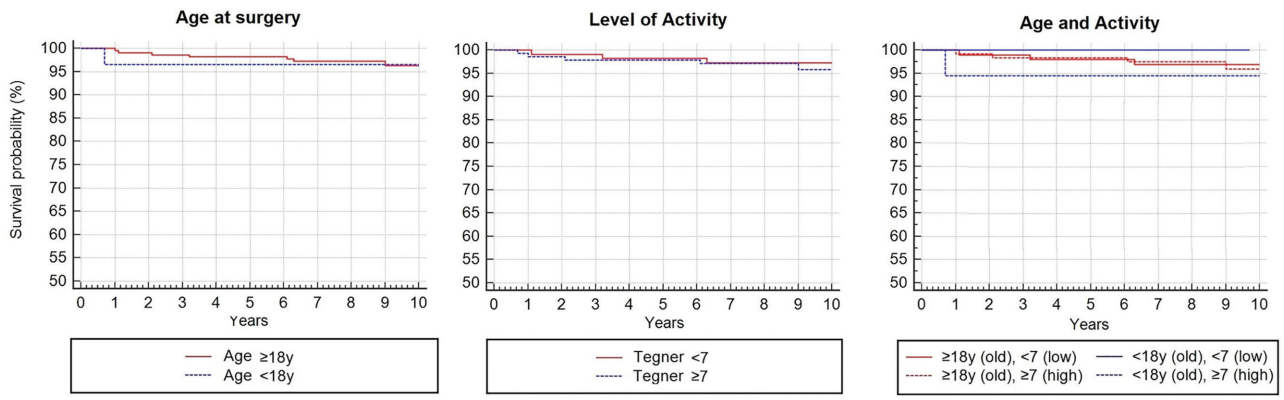

Figure 3. Survival curves of ipsilateral anterior cruciate ligament (ACL) revision. Age and activity level were not found to be risk factors.

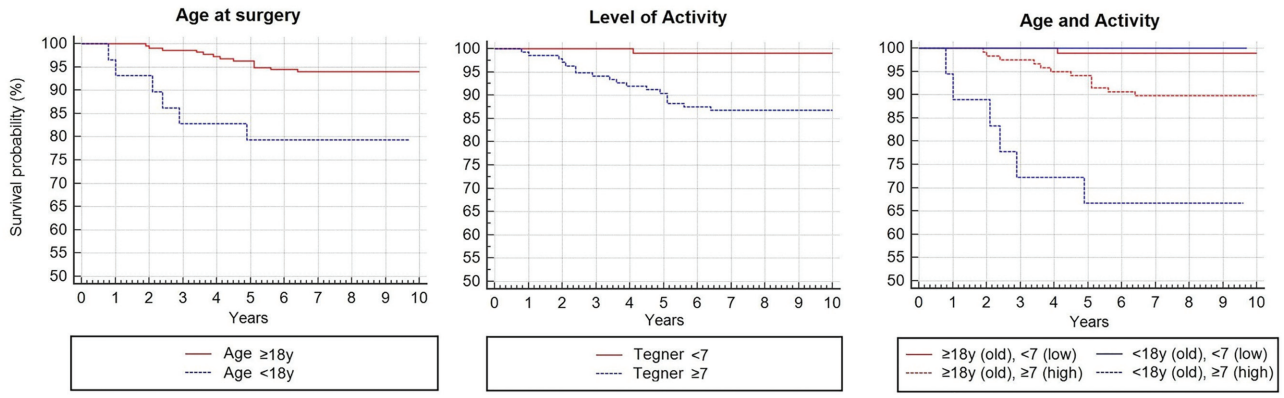

Figure 4. Survival curves from contralateral anterior cruciate ligament (ACL) reconstruction. Age and activity level were found to be significant risk factors; patients who had a combination of the 2 factors had the lowest survival rate.

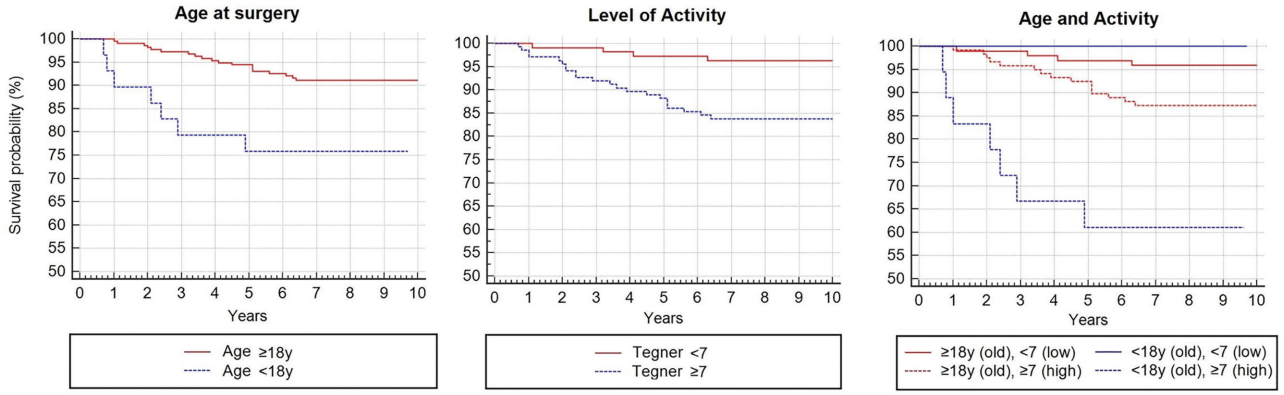

Figure 5. Survival curves of a second anterior cruciate ligament (ACL) injury. Age and activity level were found to be significant risk factors; patients who had a combination of the 2 factors had the lowest survival rate.

was present compared with ipsilateral ACL reruptures 10 years after ACL reconstruction with hamstring tendon grafts, thus rejecting the initial hypothesis. However, the second hypothesis was confirmed because activity level and young age at surgery were found to be relevant predictors of a second ACL injury. The long-term evaluation with a minimum of 10 years' follow-up, the consistent surgical technique from a single center, and the high follow-up rate, similar to

TABLE 3
Risk of Ipsilateral or Contralateral ACL Injury[a]

Time Points	Ipsilateral ACL Revision, %	Contralateral ACL Reconstruction, %	Hazard Ratio (95% CI)	*P* Value
All patients (N = 244)				
2 y	1.2	1.6	1.3 (0.3-5.9)	.7044
4 y	2.0	4.5	2.2 (0.8-6.2)	.1381
6 y	2.0	7.4	3.6 (1.4-9.6)	.0100[b]
8 y	2.9	7.8	2.7 (1.1-6.0)	.0210[b]
10 y	3.4	7.8	2.4 (1.1-5.3)	.0356[b]
Log-rank test				.0287[b]
Male sex (n = 186)				
2 y	1.6	2.1	1.3 (0.3-5.9)	.7038
4 y	2.1	5.3	2.5 (0.8-7.8)	.1157
6 y	2.1	9.1	4.2 (1.5-12.4)	.0080[b]
8 y	2.6	9.8	3.6 (1.4-9.5)	.0096[b]
10 y	3.2	9.8	3.0 (1.2-7.4)	.0169[b]
Log-rank test				.0185[b]
Tegner level ≥7				
2 y	1.5	2.9	2.0 (0.4-10.7)	.4189
4 y	2.9	8.1	2.7 (0.9-8.4)	.0765
6 y	2.9	12.5	4.2 (1.5-12.3)	.0076[b]
8 y	3.7	13.2	3.6 (1.4-9.4)	.0091[b]
10 y	4.4	13.2	3.0 (1.2-7.3)	.0159[b]
Log-rank test				.0047[b]

[a]The risk of a contralateral ACL injury was significantly higher than that of ipsilateral graft failure starting from the sixth postoperative year; the risk was 4-fold higher in male patients and in patients with a high activity level (Tegner ≥7). ACL, anterior cruciate ligament.
[b]Statistically significant.

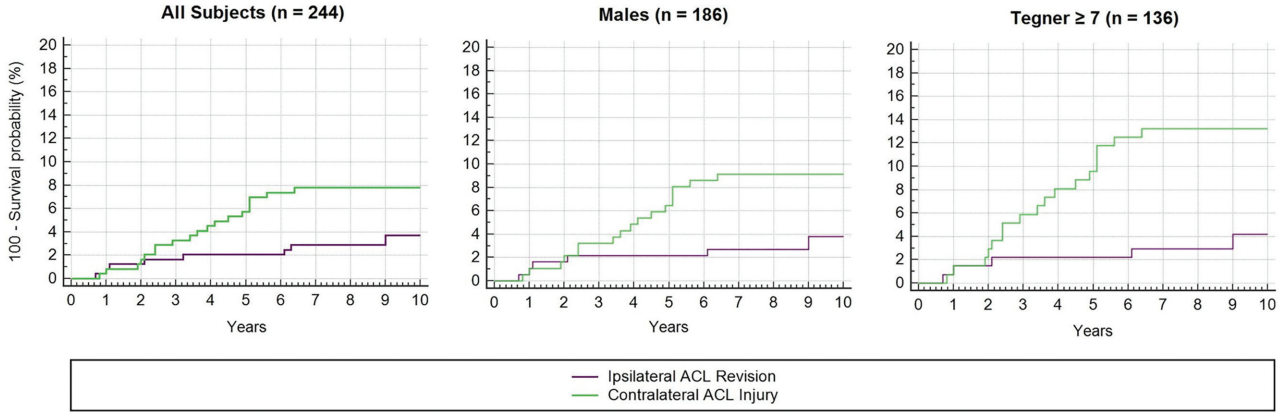

Figure 6. Survival curves of ipsilateral anterior cruciate ligament (ACL) revision versus contralateral ACL revision. For all patients, the risk of a contralateral ACL injury was significantly higher than that of ipsilateral graft failure starting from the sixth postoperative year; male patients and patients with a high activity level (Tegner ≥7) were found to be at a higher risk.

the Multicenter Orthopaedic Outcomes Network (MOON) ACL cohort,[12] represent the study's major strengths, which make the findings relevant for sports medicine clinicians.

The role of contralateral ACL reconstruction and second ACL injuries represents a topic of research that is receiving increasing interest,[11,17,23-26] and the present study contributes by highlighting the higher risk of injuries in the unaffected knee compared with the one involved in the primary surgery. This finding seems in contrast with those reported in ACL registries, where similar rates were found within 2 years after primary ACL reconstruction.[28] However, the longer follow-up of the present study could have been relevant to unmask the higher risk of contralateral injuries compared with ipsilateral reruptures. In fact, despite not performing an in-depth analysis of second injuries, Shelbourne and Gray[19] grossly reported 5.8% of patients suffering ACL failure and 11.5% undergoing contralateral ACL reconstruction in their series, thus supporting the finding of the present study. This conspicuous involvement of the contralateral knee could be due to

several reasons: first of all, there could be a tendency of the uninjured knee to protect the injured one and thus bear more loads[4]; second, the loss of proprioception due to an ACL rupture can expose either knee to further injuries[27]; and finally, unbalanced rehabilitation could also play a relevant role because focusing on the operated knee could leave the contralateral one at a higher risk.[16] Interestingly, a recent meta-analysis found a 235% increase in the risk of contralateral injuries in those patients passing test batteries for return to sports after primary ACL reconstruction.[25]

The role of the graft and surgical technique is worthy of reflection, as most of the previous studies that highlighted more contralateral injuries than ipsilateral failure were performed using the bone–patellar tendon–bone.[13,15,19] However, comparable rates between ipsilateral and contralateral reinjuries were reported with hamstring tendons.[8,15] In the present study, the pattern of second injuries appears more similar to what is reported for the bone–patellar tendon–bone rather than for the hamstring tendon. This could be because all patients were treated with combined ACL and lateral plasty, which has been proven to improve the control of rotational laxity,[2,5] thus possibly influencing the long-term survival of the reconstruction procedure. For this reason, it is also important to emphasize that results of the present study cannot be generalized because they are related to a rare technique of ACL reconstruction, however, the high rate of contralateral injuries could be interpreted in a broader perspective.

The other important finding is that according to the multivariate analysis performed in our study, predictors of second injuries were found, thus suggesting care when treating young athletes participating in demanding sports activities. This subgroup of patients could incur a second ACL injury in up to 40% of cases, delineating a dramatic scenario. Webster et al[24] described a 29% rate of second injuries in either knee in patients younger than 20 years at a shorter follow-up of 4.8 years. It is probable that the continuous exposure to the injury risk typical of young and active patients would cause a linear increase of second injuries with the passing of time.

The present study has several limitations. First, the retrospective nature of the design could have affected the injury rates; however, the inclusion of a homogeneous series of consecutive patients who underwent surgery at a single center within a limited amount of time should mitigate this limitation. Moreover, although a 15.1% rate of loss to follow-up appears high, it must be pointed out that this represents a rather high standard for a 10-year follow-up study; in fact, the 10-year follow-up rate of the MOON cohort has been reported to be 82%.[12] Another relevant limitation is the lack of objective evaluations, which could have underestimated the second injury rate. In fact, subclinical failure cases not requiring ACL reconstruction could have been undetected because of decreasing sports activities. However, all patients were carefully interviewed to confirm further injuries and surgical procedures, and in the case of trauma, they were invited to the clinic for a clinical examination to confirm ACL graft integrity. Unavoidably, this limitation is common in many studies with a similar design[23,24] and with registries.[9,20,21,28] On the other hand, the use of a clear outcome such as ipsilateral ACL revision or contralateral ACL reconstruction leaves no room for interpretation and makes the results of this study comparable with other series with similar populations. Finally, the limited number of patients belonging to specific subgroups (eg, young and active) could have impaired the calculation of HRs and survival rates. For this reason, further studies should be conducted on this specific patient population to confirm the up to 40% rate of second ACL injuries. It should also be noted that no information was provided regarding return to sports after ACL reconstruction, which could have surely affected the injury risk. However, the aim of the study was to identify predictors of injuries; therefore, we included in the analysis only the preoperative variables that influenced the injury risk at the moment of ACL reconstruction and did not consider variables that could have changed during the follow-up. Moreover, there is controversy at present regarding the definition of return to sports. By including the preoperative Tegner level in the analysis, the role of sports activity is evaluated to some extent because a higher preoperative activity level is usually translated into a higher rate of return to sports.[1,7]

The clinical relevance of the finding reported in the present study is that accurate rehabilitation of both knees should be pursued, considering the higher incidence of contralateral ruptures. In fact, any further ACL injury represents a devastating outcome for patients. In particular, care should be taken with patients younger than 18 years and those involved in high-demand sports activities because their reinjury rates are far higher than those of older and less active patients, especially regarding the contralateral healthy knee.

CONCLUSION

According to the results of the present study, a second ACL injury after primary reconstruction occurred in about 10.7% of the patients at 10 years' follow-up; the risk related to contralateral ACL reconstruction was significantly higher than that of ipsilateral ACL revision, especially in young and active patients in which the failure rate could reach 40%. This knowledge might help clinicians in delineating improved rehabilitation protocols and not neglecting the "healthy" knee, with the aim of preventing further injuries.

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